Ashutosh Jha

PhD Student, Machine Learning
ashutoshjha.net | Google Scholar

Email: a.jha@tu-braunschweig.de
Braunschweig, Germany
Languages: English (C1), Deutsch (B1)

Education

PhD in Machine Learning — Sep 2026 - Present
Institute of Artificial Intelligence, TU Braunschweig, Germany
Advisor: Prof. Dr. Michel Besserve
Causal and probabilistic representation learning for complex systems.

M.Sc. Quantitative Data Science Methods, Machine Learning & Econometrics — Oct 2023 - Jun 2026
Eberhard Karls Universität Tübingen, Germany — *Grade: 1.48/5.0 (1.0 = best)*
Thesis: Optimal Transport in linear Independent Component Analysis, at the Max Planck Institute for Intelligent Systems.
Advisors: Dr. Simon Buchholz, Prof. Dr. Michel Besserve, Prof. Dr. Joachim Grammig

Erasmus Exchange Semester — Financial Economics and Machine Learning — Feb 2025 - Jun 2025
Katholieke Universiteit Leuven, Belgium — *Grade: 1.5/5.0 (1.0 = best)*

B.E. Electrical and Electronics Engineering — Aug 2016 - May 2020
Birla Institute of Technology and Science, Pilani, Goa Campus, India — *Grade: 8.12/10*

Publications

Linear Independent Component Analysis via Optimal Transport
Ashutosh Jha, Michel Besserve, Simon Buchholz.
9th Workshop on Tractable Probabilistic Modeling (TPM), UAI 2026, Amsterdam.
arXiv:2607.14081 | OpenReview | PDF

- Measures non-Gaussianity by the squared Wasserstein distance to a standard Gaussian rather than by proxy contrast functions, with a proof that this objective is maximised exactly when a projection recovers an independent component. The resulting OT-ICA algorithm outperforms proxy-based methods on simulated data and is validated on EEG artifact removal and econometric price discovery.

Research Experience

Student Research Assistant — Perceiving Systems — *Nov 2025 - Jun 2026; Jul 2024 - Jun 2025*
at Max Planck Institute for Intelligent Systems, Tübingen, Germany

- Extended the NICE toolbox to process non-verbal communication data by integrating computer-vision detectors for pose estimation, emotion detection, and head orientation. Tools: Python, PyTorch.
- Containerised the application and managed CI/CD pipelines to make experimental models reproducible across machines. Tools: Docker, Apptainer, GitLab CI/CD, Hugging Face, Linux.

Quantitative Analyst Intern — Financial Stability — *Jul 2025 - Sep 2025*
at Deutsche Bundesbank, Frankfurt, Germany

- Modelled investment-fund portfolio vulnerabilities against market, monetary-policy, and geopolitical shocks by stress-testing econometric, probabilistic decision-tree, and foundation tabular (TabPFN) time-series models. Tools: Python, SQL, Git.

Industry Experience

Software Engineer, HSBC Holdings PLC, Pune, India — *Sep 2020 - Sep 2023*
Cloud economics and FinOps: statistical cost-benefit models ($500K savings, 1.5× vendor discounts) and real-time ETL pipelines at 4× lower latency. Python, SQL, GCP, AWS, Jenkins.

Software Engineering Intern, Nvidia Corporation, Bengaluru, India — *Aug 2019 - Dec 2019*
Real-time visualisation tool for root-cause analysis of SoC CPU boot-sequence verification. Python, Linux, shell scripting.

Selected Projects

TuebiFit — Full-Stack Agentic Application (Heilbronn Hackathon 2026) *Mar 2026*
Git Repository — LLM agents and Model Context Protocol (MCP) servers orchestrated with LangChain and LangGraph, with real-time pose estimation (MediaPipe), deployed on Google Cloud Run.

Probabilistic Asset Pricing, Master Seminar, Universität Tübingen *Nov 2024 - Jan 2025*
Git Repository — Bayesian asset pricing using asset-pricing theory as a prior and a state-space representation (Kalman filters) to estimate posterior prices, anomalies, and risk premia.

Awards

Best Pitch & Co-creation Prize, Neckarthon, Startup Center Tübingen *Nov 2024*
Dual prize for "Tue-bi-smart", an ML-enabled transport prototype for Stadtwerke Tübingen.

Prime Minister's Scholarship Scheme, Government of India *2017*
Awarded for academic excellence during undergraduate studies.

Technical Skills

Machine Learning & Research: *PyTorch, deep learning, probabilistic & Bayesian modelling, optimal transport, causal inference, time-series forecasting, LLM agents (LangChain, LangGraph), Hugging Face, TabPFN.*
Programming & Databases: *Python, SQL, R, MATLAB, VBA.*
MLOps & Tools: *Git, Docker, Apptainer, CI/CD (GitLab CI/CD, Jenkins), HTCondor, Linux, LaTeX.*
Cloud & BI: *GCP (Compute, BigQuery, Cloud Run), AWS (Lambda, S3), Power BI, MS Excel.*
Relevant Coursework: *Machine Learning, Deep Learning, Probabilistic Machine Learning, Optimization, Bayesian Modelling, Advanced Time Series Analysis, Machine Learning in Econometrics, Financial Economics.*